

RE07 Global Infrastructure Investment and Management

Week 4

Infrastructure Financing and Deal Structuring

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Guest session: International Finance Corporation

What this week does

Weeks 2 and 3 established who decides and who pays. This week is about the instrument that connects them: a contract, or rather a network of contracts, that converts a physical asset into a stream of payments reliable enough to borrow against. The syllabus asks specifically for a comparison of project finance with corporate finance and for cases from both advanced and developing economies, and the week is built around that structure.

A	Project finance against corporate finance	35 min
B	The special purpose vehicle and its contracts	35 min
C	Risk identification and allocation	40 min
D	Sources of funds and the capital stack	40 min
E	Procurement routes and how the deal is won	30 min
F	Public support, guarantees and credit enhancement	30 min
G	Two cases: Murcia and Odebrecht	35 min

What you already have

Where this week draws on modules you have taken, or are taking now. Nothing below is assumed: each idea is stated in full before the module is named.

Your programme	Where it was taught	What it gives this week
Real Estate Finance	RE01 sessions 2 to 4, corporate finance, capital structure and the weighted average cost of capital; sessions 6 and 7, financial and real options. RE03 weeks 2, 6 and 7, risk, feasibility and funding the project.	Project finance is capital structure theory at its limit: a single-asset firm with very high leverage and a contractual network standing in for covenants. RE03 follows the same sequence for a building.
Planning, Growth and Regeneration	PGR01 Lent lecture 9, planning for growth and land value capture. PGR02 Michaelmas week 8, land value capture. PGR07 lecture 3, financing mechanisms behind regeneration. PGR04 lecture 4, rent-seeking.	Where the money comes from when there is no user charge, and why a contract signed after competition is renegotiated afterwards.
Environmental Policy	EP02, elements of the theory of environmental policy I to III; environmental policy under asymmetric information; policy under risk and the use of liability rules. EP06 lectures 7 and 8, push and pull instruments.	Choosing between procurement, concession and availability payment is the same problem as choosing between a tax, a quantity instrument and a property right: which instrument delivers the outcome at least cost given what the principal cannot observe.
Everyone	RM01 lectures 1 to 4, regression and model specification.	A financial model is a set of assumptions with arithmetic attached. The habits of specification, sensitivity and diagnostic testing transfer directly to block D and to week 6.

Block A

Project finance against corporate finance

What is actually different, and why the difference is worth its cost.

Indicative length 35 minutes

A definition worth being precise about

Project finance is the financing of a single-purpose entity on the strength of the cash flow that entity is expected to generate, with limited or no recourse to the sponsors who created it. The lenders look to the project rather than to a balance sheet, and their security is the project itself together with the contracts that produce its revenue.

That definition contains the whole subject. Single purpose means no other business to contaminate or to support. Cash flow means the asset must be contracted or regulated well enough to be forecast. Limited recourse means the sponsor's loss is capped at its equity, which is why the structure exists at all.

The comparison the syllabus asks for

	Corporate finance	Project finance
Borrower	An operating company with many assets	A special purpose vehicle with one
Basis of credit	The balance sheet and historic earnings	Forecast cash flow from a single asset
Leverage	Moderate, constrained by the business risk	High, often seventy per cent or more
Security	General claims, sometimes secured	Project assets, contracts, accounts and shares
Control of cash	Management discretion	A payment waterfall with lock-up tests
Recourse to owners	Full, within the company	Limited, and defined by contract
Cost of arranging	Low	Very high, in both fees and time
Where risk sits	Diversified across the firm	Allocated by contract to named parties

Why a sponsor would accept the cost

- Risk containment. A failure is confined to the vehicle and does not reach the sponsor's other businesses, which matters most where a single project is large relative to the sponsor.
- Higher leverage. Contracted cash flow supports debt that a corporate balance sheet could not, which raises the return on the equity committed.
- Off balance sheet treatment, where the accounting permits it, though this is far more constrained than it once was and is a weak reason on its own.
- Risk sharing with partners. A consortium can undertake a project none of its members would attempt alone.
- Discipline. High leverage and a tight covenant package constrain management discretion, which is the agency argument Jensen made about debt generally.
- Against all of this sits the cost: arranging a project financing takes eighteen months and a great deal of money, and it is worth it only for large, long-lived, contractually stable assets.

The theory: separation, and the conflicts it creates

Gatti sets out the academic account. Separate incorporation avoids contamination risk in both directions: the project cannot drag down the sponsor and the sponsor's difficulties cannot reach the project. High leverage and a single asset then create a sharp conflict between sponsors and lenders, because equity holders in a highly geared vehicle have an incentive to take risk whose upside they capture and whose downside falls on lenders.

Every structural feature students meet this week is a response to that conflict: the covenant package, the reserve accounts, the lock-up tests, the restrictions on distributions, and the security over shares that lets lenders take the project away.

The empirical validation, by the author of the set text

2,000+

Project finance loans used to test whether the network of non-financial contracts around a vehicle measurably affects its financing.

Corielli, Gatti and Steffanoni ask whether the contractual architecture is doing real work or is merely legal ritual. They find it is doing real work.

A well-designed set of contracts that pre-commits the actions of construction contractors, operators, suppliers and offtakers reduces loan spreads and supports higher debt-to-equity ratios. In other words, lenders read the contracts, price them, and lend more against a project whose counterparties are properly bound.

For students this converts an apparently descriptive topic into a quantitative one. Contract design has a price, and the price is observable.

Where the technique came from

Project finance is old. Lending against the output of a specific venture, with recourse limited to that venture, was used for maritime trade and for mining long before it acquired the name. Its modern form developed in North Sea oil and in independent power generation, where the assets were large, the output was contracted to a creditworthy buyer, and the sponsors were unwilling to place the whole exposure on their balance sheets.

That history explains the shape of the technique. It was built for assets with a single identifiable output sold under a long contract, which is why it fits power and process plant naturally, transport with more difficulty, and social infrastructure only once availability payments were invented to manufacture the missing contract.

Where it is used, and why the pattern looks as it does

Sector	Fit with the technique	Why
Power generation	Very strong	A single measurable output sold to one buyer under a long contract
Renewables	Very strong	As above, with a subsidy or contract for difference supplying the offtake
Oil, gas and process	Strong	Large, contracted, and sponsored by counterparties with the capacity to support
Transport with availability payments	Strong	The state manufactures the contracted revenue that demand risk would deny
Toll roads and airports	Moderate	Demand risk sits with the project, so leverage is lower and pricing wider
Water and social infrastructure	Moderate	Contracted, but with a public counterparty and heavy political exposure
Digital and data centres	Growing	Contracted to strong tenants, but with a technology cycle far shorter than the debt

When not to use project finance

- Where the project is small relative to the sponsor, because the arranging cost cannot be recovered.
- Where revenue cannot be contracted or regulated, because there is nothing to size debt against.
- Where the technology is unproven, since lenders will not take technology risk at any price and the sponsor ends up guaranteeing it anyway.
- Where speed matters, because eighteen months to financial close is normal and cannot easily be compressed.
- Where the sponsor has cheap balance sheet capacity and no reason to ring-fence, in which case corporate debt is simply cheaper.
- The honest summary is that project finance is expensive machinery justified only by scale, contractual stability and a genuine need to contain risk.

Block A discussion

Seminar questions

1. A utility with a strong balance sheet is building a plant that is small relative to its assets. Should it use project finance? What changes your answer?
2. Project finance carries high arranging costs and takes far longer. Under what conditions does the leverage benefit outweigh that, and how would you test it?
3. Corielli and colleagues find that contract quality moves loan spreads. What does that imply about who should be in the room when the project agreements are drafted?

Block B

The special purpose vehicle and its contracts

One company, no employees, and a web of agreements that produce a bankable cash flow.

Indicative length 35 minutes

A company built to be predictable

The special purpose vehicle exists to hold one project and nothing else. It typically has no employees, no premises and no operating capability of its own. Everything it needs is bought under a contract with a party better placed to provide it: construction from a contractor, operation from an operator, inputs from a supplier, and revenue from an offtaker or from users.

The purpose of that arrangement is not efficiency in any ordinary sense. It is predictability. Each contract removes a source of variance from the vehicle's cash flow by transferring it to somebody who can bear it, and what is left is a stream stable enough for lenders to advance seventy per cent of the cost against.

The contracts, and what each one removes

Contract	Counterparty	Variance it removes
Concession or implementation agreement	The contracting authority	The right to build and operate, and the terms on which it can be taken away
Construction contract, usually turnkey	The contractor	Cost overrun, delay, and performance on completion
Operation and maintenance agreement	The operator	Operating cost and availability during the operating period
Offtake or purchase agreement	The offtaker	Price and volume risk on what the project produces
Input supply agreement	The supplier	Price, quantity and quality of what the project consumes
Shareholders agreement	The sponsors	Governance, funding obligations and what happens when they disagree
Financing agreements	The lenders	Drawdown, repayment, covenants and enforcement
Direct agreements	Lenders with each key counterparty	The risk that a key contract terminates before lenders can intervene

The turnkey construction contract does the heaviest lifting

A turnkey, or engineering procurement and construction, contract places single-point responsibility on one contractor for design, procurement, construction and commissioning, at a fixed price and by a fixed date, with liquidated damages for delay and performance shortfall and a guarantee behind them.

That is expensive. A contractor asked to carry that risk prices it, and the premium is real. Lenders insist on it anyway, because the alternative is that the vehicle carries interface risk between separate design and construction packages, and a vehicle with no employees cannot manage interfaces. This is the clearest example in the module of paying a party to bear a risk because it can manage it and you cannot.

Availability, offtake and the shape of revenue

- A take-or-pay offtake obliges the buyer to pay whether or not it takes the output, which removes volume risk entirely and is the strongest form.
- A capacity and energy structure splits payment into a fixed capacity charge, which covers debt service and fixed costs, and a variable charge for what is used.
- An availability payment pays for the asset being available to a standard, with deductions for unavailability and poor performance, and no volume exposure at all.
- A merchant structure has no contract and the project sells into a market, which is equity risk and is very rarely financed at project finance leverage.
- The lender's question in every case is the same: what is the minimum cash the project receives in a bad year, and does it cover debt service.

Step-in, and why lenders want the project rather than the assets

If a project fails, foreclosing on a half-built tunnel is worthless. What lenders want is the ability to keep the project alive under new management, which requires two things: security over the shares in the vehicle, so they can replace its owners without disturbing its contracts, and direct agreements with each key counterparty, obliging that party to continue and to give lenders a period to cure a default before terminating.

This is why recovery rates on project finance debt are high. The lenders are not selling an asset into a distressed market; they are taking over a going concern with contracted revenue and finding somebody else to run it.

Who sponsors a project, and what each wants

Industrial sponsor

Has a core business the project serves: a utility, an energy company, a manufacturer securing supply. Wants the asset to work, and will support it beyond the contract if it must.

Contractor sponsor

Builds the project and takes equity to win the construction contract. Its return is substantially earned in construction, which is a conflict lenders watch closely.

Public sponsor

A state entity pursuing a welfare objective, often holding a minority stake to retain influence and information.

Financial investor

An infrastructure fund or institution with no operating role, seeking a return and an exit. Will not support the project beyond its contractual obligation.

Operator sponsor

Runs the asset and takes equity to secure the operating contract. Aligned during operation, and a weak counterparty in construction.

Why the mix matters

Lenders read the shareholder register as a statement about who will still be there when something goes wrong.

The advisers around the vehicle

A project financing requires legal advisers for the sponsors, for the lenders and often for the authority, an independent engineer reporting to lenders on technical matters, an insurance adviser, a model auditor, a market consultant to validate demand or resource assumptions, and financial advisers on both sides.

The cost is substantial and it is not decoration. Each adviser exists to give one party independent assurance about a matter it cannot verify itself, which is a direct response to the information asymmetry that makes the transaction difficult. Week 5 asks what that assurance is actually worth, and whether the fee corresponds to it.

The security package, and what lenders actually take

- A pledge over the shares in the project company, which allows lenders to replace the owners without disturbing the contracts.
- Security over the project's receivables, so revenue is captured at source.
- Security over the project accounts, which is what makes the waterfall enforceable.
- Mortgages or equivalent over the project property, more useful as a blocking right than as a realisable asset.
- Assignment of the project agreements and of insurance proceeds.
- Direct agreements with each key counterparty, which convert all of the above into a workable ability to step in and continue.

Conditions precedent, and why nothing is effective until everything is

Financial close occurs when a long list of conditions is satisfied simultaneously: contracts executed, consents obtained, land secured, insurance bound, security perfected, legal opinions delivered, equity commitments in place, and the base case model agreed and audited.

The list is long because every party is unwilling to be committed while any other is not. That produces a characteristic pattern in which the transaction appears stalled for months and then completes in a single day, and it is why a slipping condition precedent, however minor, can hold an entire financing.

Block B discussion

Seminar questions

1. A project company has no employees. What does that imply about who actually bears operational risk, and about what the sponsor is being paid for?
2. A turnkey contract is more expensive than separate design and build packages. Who benefits from the premium, and would you pay it if you were funding the project with your own money?
3. Rank the eight contracts by how much a lender would care if it were weak. Defend the top two.

Block C

Risk identification and allocation

What can go wrong, who should carry it, and what happens when the answer is nobody.

Indicative length 40 minutes

Risks by project phase

Phase	Principal risks	Usual bearer
Pre-completion	Planning and consent, land, technology, construction cost and delay, performance on commissioning	Contractor, with sponsor support and, for consent, the authority
Post-completion	Operating cost, availability, maintenance and lifecycle, input supply, demand and price	Operator, offtaker, supplier, and the party best able to influence demand
Throughout	Interest rate, exchange rate, inflation, environmental and social, regulatory and political, force majeure, counterparty credit	Hedged, shared, or retained by the authority where it is created by the state

The principle, and how often it is ignored

A risk should sit with the party best able to influence its likelihood or its consequences, and where nobody can influence it, with the party that can absorb it at least cost. That is the whole of risk allocation theory and it is easy to state.

It is ignored constantly, in a predictable direction. Public authorities transfer risks the private party cannot manage, principally demand and policy risk, because transfer looks like prudence. The private party prices the risk, or fails to and renegotiates later. Either way the authority pays, and in the second case it pays at the worst moment.

What practitioners actually think

Delphi

Method used by Ke and colleagues to establish preferred risk allocation in Chinese public private partnership projects.

Rather than assert a risk matrix, the authors ran a two-round Delphi survey with experienced practitioners and recorded where consensus lay.

The pattern is clear. The public sector is expected to take sole responsibility for expropriation and nationalisation, and the majority of responsibility for a further twelve risks connected to government and the actions of officials. Fourteen risks that neither side can control alone are preferred to be shared equally. The private sector is expected to carry the majority of ten project-level risks.

The value for teaching is that students get an empirical risk matrix to argue with rather than a normative one to accept.

The risks that are hardest to place



- Demand. Neither party controls it in most transport projects, so the question is who absorbs it more cheaply, and a diversified state usually can.
- Change in law. Created by the state, so in principle the state's, but a general change affecting all businesses is not obviously the project's to be compensated for.
- Land and consent. Usually the authority's, because only the authority has compulsory powers, and a project that begins before land is secured has an unmanaged exposure.
- Inflation. Shared through indexation, and the detail of which index and what share of revenue matters more than the principle.
- Currency. Rarely allocable at a sensible price in the markets that need it most, which is the week 3 barrier appearing as a contract problem.
- Force majeure and its modern cousins, including pandemic and extreme weather, which are increasingly negotiated rather than boilerplate.

Risk allocation is priced, and the price is visible



Every transfer has a cost. A contractor asked to carry ground conditions risk prices it, and the price appears in the construction sum. An offtaker asked to take volume risk prices it, and the price appears in the tariff. A lender facing residual risk prices it in the margin and the cover ratio it requires.

The practical implication is that the risk matrix and the financial model are the same document seen from two angles. A student who understands this can look at a tariff and infer the risk allocation behind it, which is a genuinely useful skill and one the group project will reward.

Environmental and social risk is a credit risk



Lenders to infrastructure apply environmental and social standards, most commonly the Equator Principles and the IFC Performance Standards, which cover impact assessment, labour, resource efficiency, community health, land acquisition and involuntary resettlement, biodiversity, indigenous peoples and cultural heritage.

The reason is not primarily ethical. A project that has not managed its social impact faces delay, litigation, protest and, in the limit, withdrawal of consent, all of which are credit events. Land acquisition and resettlement in particular have stopped more projects than technical failure has. For the environmental policy students this is the point at which the standards they study become a financing condition rather than an aspiration.

Political risk, and its forms

Form	What happens	Usual mitigation
Expropriation	The asset is taken, with or without compensation	Political risk insurance; multilateral participation
Creeping expropriation	Tariffs frozen, permits withheld, taxes changed until the return disappears	Contractual change in law protection; arbitration
Transfer and convertibility	Revenue cannot be converted or remitted	Offshore accounts; insurance; local currency debt
Breach of contract	The authority does not perform and cannot be compelled	Direct agreements; multilateral lenders; arbitration awards
Political violence	War, civil disturbance, terrorism	Insurance, where available at a price
Non-honouring of a guarantee	A sovereign undertaking is simply not met	Multilateral guarantee; preferred creditor effect

Insurance as a risk transfer instrument



- Construction all risks and delay in start-up cover during the build, which is what protects debt service if completion slips for an insured reason.
- Operational property damage and business interruption cover once the asset is running.
- Third party liability, and environmental liability where available.
- Political risk insurance for the exposures on the previous slide, from public agencies and from a private market.
- Insurance is a transfer to a party that pools the risk, which is the efficient bearer principle in its purest form, and lenders specify the cover as a condition rather than leaving it to the sponsor.
- What insurance cannot cover is the interesting residual: demand, policy, and the expiry of a contract.

Force majeure and change in law

Force majeure covers events outside the control of either party that prevent performance. The drafting question is not whether to include it but what it does: suspend obligations, extend the concession, trigger compensation, or permit termination, and after how long.

Change in law is treated separately because it is not outside the control of both parties. The state creates it, so a general change affecting all businesses is usually left with the project while a discriminatory change specific to the project or its sector is compensated. That distinction is easy to state and endlessly argued over, and it is where a great many disputes originate.

Block C discussion

Seminar questions

1. Allocate the risks for a greenfield toll road in a middle income country. Where would you deliberately depart from the Ke consensus, and why?
2. An authority transfers demand risk to a concessionaire that cannot influence demand. Trace what happens over the following ten years under a pessimistic traffic outturn.
3. If risk allocation is priced, is a heavily transferred structure ever cheaper for the state than a lightly transferred one? Under what conditions?

Block D

Sources of funds and the capital stack

Equity, mezzanine, senior debt and bonds, and how the stack is actually assembled.

Indicative length 40 minutes

The capital stack

Layer	Who provides it	What it is paid	Where it ranks
Senior debt	Banks, institutional lenders, bondholders	Margin over a benchmark, plus fees	First, after operating costs
Subordinated or mezzanine debt	Specialist funds, sometimes sponsors	Higher margin, sometimes with equity features	After senior, before equity
Shareholder loans	The sponsors	Interest, often for tax reasons	Behind all external debt
Equity	Sponsors and financial investors	Residual cash flow and exit value	Last
Public support	The state or a development institution	Nothing, or a below-market return	Varies by instrument, and often first-loss

Debt is sized against cash flow, not against assets

In corporate lending the question is what the borrower is worth. In project finance it is what the project can service. Debt is sized so that projected cash available for debt service exceeds scheduled debt service by a required margin in every period, under a downside case rather than the base case.

That required margin is the cover ratio, and it is the subject of week 6. For now the important point is directional: a project with more contracted revenue supports more debt at the same cost, which is why sponsors spend so much effort on the offtake, and why an uncontracted merchant project is financed at leverage a fraction of a contracted one.

The senior facilities students will meet

- The base facility, which funds the construction cost and is repaid from operating cash flow.
- A working capital facility for the operating period.
- A standby facility, drawn only if cost overruns occur, which is how completion risk is funded without over-sizing the base facility.
- A value added tax facility, bridging the timing difference between paying tax on construction and reclaiming it.
- A debt service reserve facility, which can substitute for a funded reserve account and is cheaper.
- Each exists because a project has a lumpy and predictable funding profile, which lets facilities be tailored far more finely than corporate lending allows.

Bonds against loans

A project can borrow from banks or issue bonds, and the choice turns on four things. Bonds give longer tenor and fixed rates, and access to institutional investors who cannot lend directly. Loans give flexibility: they can be drawn as construction progresses, amended when circumstances change, and waived by a small group of lenders who understand the asset.

Bonds require a rating, which means an external party must be persuaded, and they require negative carry if funds are raised before they are needed. The practical resolution in many transactions is to build with bank debt and refinance into bonds once construction risk has passed, which matches the instrument to the risk profile at each stage.

Islamic finance, briefly but properly

- Structures avoid interest and require that returns be linked to an identifiable asset or to a share in a real enterprise, which suits infrastructure unusually well.
- Debt-like contracts include murabaha, salam, istisna and ijara; participation contracts include mudarabah and musharakah.
- Sukuk are certificates representing an interest in an asset or venture rather than a debt claim, and are the principal capital markets instrument.
- For infrastructure the ijara and istisna structures map naturally onto construction and lease of a physical asset.
- The market matters practically as well as intellectually: several of the jurisdictions with the largest infrastructure programmes finance substantially in this form.

Hedging, and what remains after it

A project with floating rate debt and fixed contracted revenue has an interest rate exposure it cannot bear, so it swaps into fixed. A project with hard currency debt and local currency revenue has a currency exposure it usually cannot hedge for the tenor required. A project with an inflation-indexed tariff and nominal debt has a real exposure that varies with realised inflation.

Hedging removes the exposures for which a market exists and leaves the ones for which it does not. Those residual exposures are where the real risk sits, and they should be identified explicitly rather than assumed away, because a hedge that terminates in year seven of a twenty-five year project has not solved the problem.

When equity actually goes in

Sponsors prefer to contribute equity last, after debt has been drawn, because deferring the contribution raises the internal rate of return without changing anything real. Lenders prefer equity first, because money already spent is money that cannot be withdrawn when the project looks difficult.

The usual compromise is a combination: some equity up front, the balance committed under a letter of credit or an equity contribution agreement, drawn pro rata or at the end. Where the balance is deferred, the lenders are in effect taking sponsor credit risk for the period, which is why the credit quality of the equity contribution agreement is scrutinised as carefully as the sponsor's operating capability.

The cost of capital for a project, and why it is not the sponsor's

A common error is to discount project cash flows at the sponsor's weighted average cost of capital. The sponsor's cost of capital reflects the risk of its existing business, and the project has a different risk. In a non-recourse structure the sponsor's exposure is capped, and the project's own capital structure is far more heavily geared than the sponsor's.

The correct approach is to build the project's own cost of capital from the risk of its cash flows and its own gearing, and to recognise that gearing changes over the life of the project as debt amortises, so a single constant discount rate is an approximation whose limits should be stated. This is RE01 material applied to an entity with one asset and a declining debt profile.

Refinancing, and who gets the gain

- Once construction risk has passed, the project is a better credit and can usually borrow more cheaply and for longer than at financial close.
- Refinancing captures that improvement, and the gain can be substantial: lower margin, longer tenor, and often a release of cash to shareholders.
- In a public contract the question is who owns the gain. Early PPP contracts left it entirely with the private party, which produced windfalls that became politically damaging.
- Modern contracts share refinancing gains with the authority, typically on a defined split, and require consent for refinancings that increase the authority's termination exposure.
- The lesson generalises. A contract that does not anticipate an improvement in circumstances allocates that improvement by default, and the default is rarely the one the public side would have chosen.

Block D discussion

Seminar questions

1. A project can be financed with bank debt or with a bond. Which would you choose for a greenfield renewable project, and would your answer change once it is operating?
2. Debt is sized against a downside case. Who chooses the downside case, and what stops it from being chosen conveniently?
3. A project hedges interest rate risk for ten years on a twenty-five year concession. What has it actually achieved?

Block E

Procurement routes and how the deal is won

Auctions, negotiations, and the competition that produces the contract.

Indicative length 30 minutes

Routes to award

Route	When it works	What it costs
Open competitive tender	Standardised requirements, several credible bidders, contract can be fully specified	Long, expensive for bidders, and inflexible once launched
Restricted or prequalified tender	Complex requirements where capability must be screened first	Narrows the field, and prequalification criteria can be gamed
Competitive dialogue	The authority knows the outcome it wants but not the solution	Very long, and requires an authority capable of running it
Direct negotiation	A single credible counterparty, or genuine urgency	No price discovery, and the hardest to defend afterwards
Unsolicited proposal	The proponent has something nobody else has	No competition at all unless one is manufactured afterwards

Auctions are not always better

2009

Bajari, McMillan and Tadelis on when competitive bidding underperforms negotiation in procurement.

Using a dataset of private sector building contracts in northern California, they find that auctions perform poorly where projects are complex, contractual design is incomplete, and few bidders are available.

The mechanism is that an auction suppresses the communication between buyer and seller that a complex project needs. The buyer cannot draw on the contractor's expertise while designing, so the project is designed worse and amended expensively afterwards.

The largest and most novel infrastructure projects are exactly the ones where specification is least complete and bidders are fewest, which is where the case for open tender is weakest and where procurement rules insist on it most firmly.

What makes a tender fail

- Too few bidders, which is common in large infrastructure because bid costs run to millions and only a handful of consortia can absorb them.
- Bid costs that are not recognised, so bidders price the risk of losing into the bids they win.
- Requirements that change during the process, which resets bidder pricing and rewards those who anticipated the change.
- Evaluation on price alone, which selects the bidder most willing to be optimistic, and produces the renegotiation from week 2.
- A public sector team without the capacity to run the process, which produces delay, and delay drives bidders away.
- Market sounding is the cheapest insurance against most of these and is skipped more often than it is done.

Financial close, and what actually happens in the last month

Between preferred bidder and financial close sits a period in which the transaction is frequently at its most fragile. Conditions precedent must be satisfied, lenders complete diligence and issue final credit approvals, insurance is bound, land title is confirmed, and every document is executed in a sequence in which nothing is effective until everything is.

Between bid submission and close the market can move, and a rise in the cost of debt during that window is a risk somebody carries. Who carries it should be settled at bid stage and often is not, which is why so many transactions are repriced immediately before signature.

The bidder's economics, and why they shape the market

Bidding for a large infrastructure concession costs millions and takes a year or more, and the probability of winning against three competitors is one in four. A rational bidder therefore requires that the expected value of winning covers four bid costs, and prices accordingly.

Two consequences follow. Bid costs are paid by the winning project and ultimately by users or taxpayers, so a process with many bidders is not free. And only consortia with the balance sheet to absorb repeated losses can participate, which is why the same names appear in every shortlist. An authority that wants more competition has to reduce the cost of bidding rather than exhort more firms to try.

Designing evaluation criteria

- Price alone selects the most optimistic bidder, which is the winner's curse and then the renegotiation.
- Weighted quality and price requires the authority to define quality in advance, which is exactly the contractibility problem from Hart.
- A fixed tariff with competition on quality inverts the problem and works where the service is specifiable.
- Competition on a single financial variable, such as the required subsidy or the concession length, is transparent and easy to defend, and only works where everything else is genuinely fixed.
- Whatever the criteria, they must be published, applied as published, and capable of surviving challenge, because a procurement overturned late is more expensive than one designed slowly.

Standardisation is the cheapest reform available

Where an authority has a programme rather than a project, standardising the contract removes cost from every subsequent transaction. Bidders price a known document rather than reading a new one, legal fees fall, negotiation shortens, and the risk allocation becomes a policy decision made once rather than a negotiation conducted repeatedly by whoever is in the room.

The United Kingdom's standardised PFI contract is the best-known example and its lesson is double-edged. Standardisation delivered real savings, and it also propagated a risk allocation across hundreds of contracts, so that when the allocation turned out to be wrong it was wrong everywhere at once.

Block E discussion

Seminar questions

1. Bajari and colleagues find auctions underperform where contracts are incomplete. Public procurement rules generally require open tender for the largest contracts. How would you reconcile the two?
2. A tender attracts two bidders. Should the authority proceed, and what would you do differently next time?
3. Who should carry the risk that debt markets move between bid and financial close?

Block F

Public support, guarantees and credit enhancement

What the state can add to make a transaction work, and what each addition costs it.

Indicative length 30 minutes

Funded and contingent support

Instrument	Funded or contingent	What it changes	What it costs the state
Capital grant	Funded	Reduces the cost to be recovered from tariffs	Immediate and visible
Availability payment	Funded	Removes demand risk entirely	A long-dated commitment, recorded as such
Concessional loan	Funded	Lowers the blended cost of capital	The difference from market rate, rarely computed
Minimum revenue guarantee	Contingent	Truncates the downside for equity and lenders	A written option, usually unpriced
Partial credit guarantee	Contingent	Raises the rating of the debt	Contingent exposure to the full guaranteed amount
Political risk insurance	Contingent	Covers expropriation, transfer and contract breach	A premium, if bought commercially
Termination compensation	Contingent	Makes the transaction bankable at all	The largest exposure, and the least modelled

Credit enhancement, and what it is really doing

A partial guarantee from a highly rated institution raises the rating of a project's debt, which widens the set of investors permitted to hold it. That is its purpose: not to make the project safer in any physical sense but to move it across a threshold in somebody's mandate.

Saying it that way is not cynical. Mandate thresholds are real constraints and moving across one genuinely changes the cost and availability of capital. But it does mean the value of the enhancement should be measured as the change in the cost of funds, against the expected cost of the guarantee to the guarantor, and that comparison is made much less often than it should be.

Viability gap funding, and the discipline it requires

Many worthwhile projects cannot support their full cost from tariffs that users can pay. Viability gap funding closes that difference with a capital contribution, usually competed so that bidders compete on the smallest subsidy required.

Competing the subsidy is what makes the instrument defensible: it reveals the minimum public contribution the market will accept rather than leaving it to negotiation. Where the gap is instead set administratively and then offered, the state has no way of knowing whether it paid too much, and the answer is usually yes.

How a blended structure is actually assembled

Layer	Provider	Function
Grant for preparation	Bilateral agency or facility	Pays for the feasibility and transaction work nobody else will fund
Viability gap capital grant	The state	Reduces the cost to be recovered from users
Concessional senior or subordinated debt	Development institution	Lowers the blended cost of debt and lengthens tenor
First-loss tranche	Donor or climate fund	Absorbs the initial losses that keep commercial investors out
Guarantee	Multilateral	Raises the rating of the commercial tranche across a mandate threshold
Commercial senior debt and equity	Banks and funds	The capital the structure exists to mobilise

Questions to put to the guest

- When you decide to participate in a transaction, what are you actually optimising, and how is that traded off against loss?
- How do you price a guarantee, and does the price reflect expected loss or the concessional intent?
- What proportion of the transactions you look at fail on the project and what proportion on the jurisdiction?
- Mobilisation is estimated at around seven dollars of commercial credit per dollar of multilateral lending. Does that match your experience, and what drives the variation?
- What would you change about how development institutions and commercial investors work together?

Block F discussion

Seminar questions

1. A state gives a minimum revenue guarantee and records nothing. Value the guarantee as an option and say where it should appear in the public accounts.
2. A partial credit guarantee moves a bond from sub-investment grade to investment grade. Estimate what that is worth in basis points and compare it with the expected cost to the guarantor.
3. Should development institutions price at expected loss or below it? What follows for the market either way?

Block G

Two cases

One advanced-economy renewable project, one emerging-market project bond.

Indicative length 35 minutes

Why these two

The syllabus asks for project finance case studies in both advanced and developing economies, and Gatti supplies both in a form students can prepare from.

Case study 7 is a solar power plant in Murcia, in a regulated European market, financed with senior bank debt against a power purchase agreement, with a hedging strategy, covenants and a base case built on an energy yield assessment. Case study 3 is the Odebrecht drilling project in Brazil, financed in the capital markets with an export credit agency refund guarantee, a cash waterfall, and step-in rights, in a jurisdiction and a sector where political and counterparty risk are material.

Between them they cover almost everything in the week.

The two cases side by side

	Murcia solar	Odebrecht drilling
Jurisdiction	Spain, an OECD regulated market	Brazil, with offshore structuring
Revenue	Power purchase agreement	Charter and service agreements with a single large counterparty
Instrument	Senior bank facility	Project bond issued through a special purpose issuer
Key support	Market regulation and the offtake	Export credit agency refund guarantees, equity support agreements
Dominant risk	Energy yield, price after the contract, regulatory change	Counterparty concentration, political and country risk, construction
What it teaches	Debt sizing, covenants, hedging, sensitivity	Capital markets execution, waterfall mechanics, credit enhancement

The questions to put to both cases

- What is the minimum cash the project receives in a bad year, and what happens to the cover ratio then?
- Which party bears each of the six hardest risks from block C, and was that the efficient allocation?
- What happens when the revenue contract expires, and how much of the value rests on what happens afterwards?
- If this transaction had to be done again in a jurisdiction two notches lower, what would have to change?
- Where would you have refused to lend, and at what price would you have changed your mind?

How to read a term sheet

Section	What to look for	Why it matters
Facilities and amounts	Base, standby, working capital, and what each may fund	Tells you what the lenders think might go wrong
Tenor and repayment	Final maturity against concession length, and the profile	A tail between debt maturity and concession expiry is the lenders' cushion
Pricing	Margin, step-ups, and fees	Step-ups reveal where the lenders expect to be repaid or refinanced
Financial covenants	Cover ratio tests, lock-up levels, default levels	The distance between lock-up and default is the equity's room to manoeuvre
Accounts and waterfall	Reserve requirements and the order of payment	Where the cash actually goes and when equity sees any
Conditions precedent	What must be true before a penny is drawn	The practical critical path to financial close
Events of default	What triggers them, and what cure periods apply	How quickly a difficulty becomes a crisis

The Italy water case, and where week 6 picks it up

Gatti case study 2 is a full business plan for an integrated water system: timing, inflation, depreciation, interest, taxation, working capital, a debt service reserve, capital expenditure, sources of financing including a public grant, and an operating period covering the existing system, new supply and treatment works, and two hydroelectric plants. It closes on the economic and financial ratios.

It is the only case in the set text that runs the numbers end to end, and it is the basis for the individual coursework in week 6. Students should read it now, so that when the modelling exercise arrives the transaction is already familiar and the work is analytical rather than descriptive.

Block G discussion

Seminar questions

1. Which of the two transactions was better structured, and by what criterion are you judging?
2. The Odebrecht case relies on an export credit agency guarantee. Strip it out and re-price the transaction qualitatively. What survives?
3. Both cases were financeable. Name a project you know of that would not be, and identify the single feature that makes it unbankable.

Block H

Reserve material

Slides held back from the main sequence.

Indicative length As required

The financial model, and what week 6 will do with it

- A project financial model is the single document in which the risk allocation, the contracts and the capital structure all become numbers.
- It contains assumptions, capital expenditure, revenue, operating cost, tax, working capital, debt drawdown and repayment, reserves, and the resulting cash flows and ratios.
- It is built once and then negotiated over, and by financial close it is a contractual document rather than an analytical one.
- Week 6 builds and interrogates it. This week the point is only that everything discussed so far ends up inside it.
- Gatti's appendix walks through the structure of a real model section by section and is the basis for the individual coursework.

Candidate cases for Week 4

Case	What it illustrates	Material
Murcia solar	Contracted renewable, bank debt, covenants and hedging	Gatti case study 7
Odebrecht drilling	Emerging market project bond with export credit support	Gatti case study 3
Italy water system	A full business plan with ratios, used again in week 6	Gatti case study 2
Thames Tideway	A bespoke regulatory and financing structure with government support	Public documentation is extensive
A UK availability PPP hospital	Availability payments, deductions and refinancing	National Audit Office reports
An offshore wind farm with a contract for difference	How a policy instrument becomes a bankable revenue stream	Public auction results and company reporting

An alternative running order

Where the cohort is drawn mainly from Real Estate Finance, compress blocks A and B, since capital structure and contract are familiar territory, and give the time to blocks C and E, where the material is genuinely new.

Where the cohort is drawn mainly from Planning and Environmental Policy, invert: spend longer on A and B, treat block D as background, and use the EP02 instrument choice framing throughout block E so that procurement route selection is recognisable rather than technical.

Sources drawn on for this deck

- Gatti, S. (2024), Project Finance in Theory and Practice, chapters 1, 3, 5, 6 and 7, and case studies 2, 3 and 7.
- Delmon, J., Private Sector Investment in Infrastructure, chapters 1, 2, 3, 8, 10, 16, 20, 22 and 24.
- Abor, J. Y. and others (eds) (2025), The Routledge Handbook of Infrastructure Finance, chapters 12, 13, 17, 18 and 30.
- Weber, B., Staub-Bisang, M. and Alfen, H. W. (2016), Infrastructure as an Asset Class, chapters 6 and 7.
- APMG PPP Certification Program Guide, chapters 4 and 5.
- Journal articles as cited on each slide.

Where Week 4 leaves us

Project finance is the financing of a single-purpose entity against forecast cash flow, and every structural feature follows from that one choice.

The contracts are not documentation. They measurably move loan spreads and gearing, which makes structuring a quantitative activity.

A risk belongs with whoever can influence it, or absorb it most cheaply, and transferring it to anybody else buys an option that will be exercised.

Competition improves procurement except where contracts are incomplete and bidders are few, which is precisely the case for the largest projects.